

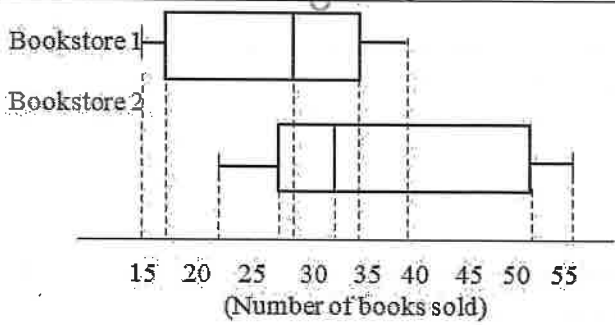
NSW INDEPENDENT TRIAL EXAMS – 2022
MATHEMATICS STANDARD YEAR 11 EXAMINATION
MARKING GUIDELINES

Section I

Question	Answer	Assessed Outcome	Band
1	A	MS-S2/MS11-10	3
2	A	MS-M1/MS11-3	3
3	B	MS-A1/MS11-10	3
4	C	MS-S1.2/MS11-7	3
5	B	MS-F1.2/MS11-6	3/4
6	B	MS-S1.1/MS11-10	3
7	C	MS-M2/MS11-10	3
8	D	MS-S2/MS11-8	4
9	B	MS-F1.3/MS11-10	4
10	D	MS-S1.2/MS11-7	4
11	A	MS-S1.1/MS11-2	4
12	D	MS-A2/MS11-6	4
13	D	MS-A1/MS11-1	4
14	C	MS-M1.2/MS11-10	4/5
15	A	MS-F1.1/MS11-9	5
16	C	MS-A1/MS11-10	4/5
17	A	MS-M1.2/MS11-4	5
18	A	MS-M1.1,1.3/MS11-3	4
19	B	MS-F1.1/MS11-6	5
20	D	MS-F1.1/MS11-9	6

Section II

Qu	Answer	Mark	Outcome Assessed	Band
21	City P has a UTC of $+3 - 7 = -4$	1	MS-M2/ MS11-10	3
22	Interest earned = $\$10\,000 \times \frac{2.2}{100} \times 4$ = \$880	1 1	MS-F1.1/ MS11-10	3
23(a)	The error in the measurement is 0.005 cm. The lower limit is $6.25 \text{ cm} - 0.005 \text{ cm}$. = 6.245 cm	1 1	MS-M1.1/ MS11-3	3
(b)	The percentage error = $\frac{0.005}{6.25} \times 100$ = 0.08%	1	MS-M1.1/ MS11-3	3
24	The phone plan cost = $\{\$40 + [\$ (1256 - 320) \div 36]\}$ /month = $\{ \$40 + 26 \}$ /month = \$66	1 1	MS-F1.3/ MS11-9	3
25(a)	$\$C = 50 - (2.5 \times 6)$ = \$35	1	MS-A2/ MS11-2	3
(b)	Let $C = 15$, then $15 = 50 - 2.5N$ $2.5N = 35$ $N = 14$ 14 books were purchased.	1 1	MS-A1/ MS11-6	3
(c)	The price per book in bookstore 1 is \$2.50. The price per book in bookstore 2 is \$4.75. The difference in price is \$2.25.	1	MS-A2/ MS11-10	3
(d)	Let $4.75N = 50$; then, $N = 10.526$ The maximum number of books that can be purchased is 10. The cost of 10 books at \$4.75 is \$47.50. So, the change from \$50 is \$2.50.	1 1	MS-A2/ MS11-10	4
26(a)	Using surface area $(S) = 3\pi r^2$ $= 3 \times \pi \times 3.25^2 \text{ cm}^2$ $= 99.549$ $= 99.5 \text{ cm}^2$ (dec pl)	1 1	MS-M1.2/ MS11-3	3
(b)	Using $S = 4\pi r^2$ $= 4 \times \pi \times 3.25^2$ $= 132.7 \text{ cm}^2$ OR $\frac{99.5}{3} \times 4 = 132.7 \text{ cm}^2$	1	MS-M1.2/ MS11-3	3

Qu	Answer	Mark	Outcome Assessed	Band
30	Area of square = 81 cm^2 Area of quadrant = $\frac{1}{4} \times \pi \times 9^2$ = 63.62 cm^2 Area of part of square outside of circle = $(81 - 63.62) \text{ cm}^2$ = 17.38 cm^2 Percentage of square's area outside of circle = $\frac{17.38}{81} \times 100$ = 21.5%	1 1 1	MS-M1.2/ MS11-4	5
31(a)	Bookstore 2 sold the highest number of books on a day (56) books.	1	MS-S1.2/1.3/ MS11-2	4
(b)	The median number of books sold at bookstore 1 is 29. The median number of books sold at bookstore 2 is 32. Hence, bookstore 2 had the highest median of 32.	1	MS-S1.2/1.3 MS11-2	3
(c)	 (1 mark for reasonable accuracy according to scale 2 marks for correct 5 number summary: 22 26 32 52 56)	3	MS-S1.2/ MS11-2	4/5
(d)	Even though bookstore 2 sold more books than bookstore 1, bookstore 1 had a smaller range, median and IQR than bookstore 2, which suggests it had more consistent sales.	1	MS-S1.2/ MS11-10	5

Qu	Answer	Mark	Outcome Assessed	Band
32(a)	Jennifer invested her money over a period of 40 years with interest paid half yearly. So, $r = (3.25\% \div 2)$ $= 1.625\%$ $= 0.01625$, and $n = 40 \times 2 = 80$ Hence, $A = \\$30\,000(1.01625)^{80}$ $= \\$108\,934.63$ Pierce invested his money over 35 years with an annual interest rate of 3.25%. So, $r = 3.25\%$ $= 0.0325$ and $n = 35$ Hence, $A = \\$30\,000(1.0325)^{35}$ $= \\$91\,891.08$ Jennifer has $\\$108\,934.63 - \\$91\,891.08$ more than Pierce. i.e., $\\$17\,043.55$	1 1 1 1	MS-F1.1/ MS11-5	4/5
32(b)	The results suggest than the earlier an investment is undertaken, the more interest will be earned and the more the accumulated amount will be. Pierce invested the same amount of money 5 years later than Jennifer, with the same interest rate paid annually (rather than half yearly) and has \$17 043.55 less in his account	1	MS-F1.1/ MS11-10	4
33	The commission of \$660 is 12% of the amount of sales over \$1500. So, the sales over \$1500 $= \frac{660}{12} \times 100$ $= \\$5\,500$ Total sales = \$1500 + \$5 500 $= \\$7000$	1 1 1	MS-F1.2/ MS11-10	4/5

Qu	Answer	Mark	Outcome Assessed	Band
34(a)	Without the outlier, the distribution of the data is symmetric.	1	MS-S1.2/ MS11-2	3/4
(b)	With the outlier, the distribution would be positively skewed.	1	MS-S1.2/ MS11-2	4/5
(c)	Using the outlier in the analysis of the data has the effect of increasing the mean giving a misleading representation of the data. In this case, the median would be a better measure to use to give a more accurate representation.	1	MS-S1.2/ MS11-10	4/5
35	The volume of water falling on the roof: $= (0.015 \times 18.5 \times 12.5) \text{ m}^3$ $= 3.46875 \text{ m}^3$ $= 3468.75 \text{ L}$ Hence, 468.75 L overflows from the tank of 3000 L.	1 1 1	MS-M1.1/ MS11-3	4/5
36(a)	From the graph, when $t = 6$ (min), V (tonnes) = 1.5	1	MS-A2/ MS11-2	3
(b)	The gradient of the line AB gives the rate of flow of wheat from the silo. Now the gradient (AB) = $\frac{2 - 1.5}{6 - 2}$ $= \frac{0.5}{4}$ $= 0.125$ tonnes This is equivalent to 125 kg/min.	1	MS-A2/ MS11-10	4/5
(c)	Let $V = -0.125t + A$ (note gradient is negative since the silo is emptying) Using $V = 1.5$, $t = 6$, from (a), $1.5 = (-0.125 \times 6) + A$ $A = 2.25$ So, $V = -0.125t + 2.25$	1 1	MS-A2/ MS11-9	5
(d)	Let $V = 0$ in $V = -0.125t + 2.25$ Then $0 = -0.125t + 2.25$ $0.125t = 2.25$ $t = 18$ (minutes) (the time it takes for the silo to empty)	1 1	MS112/ MS11-10	4/5

Qu	Answer	Mark	Outcome Assessed	Band
39(a)	$TQ = PQ - SR$ $= l_1 - l_2$	1	MS-M1.2/ MS11-10	3
(b)	Area of trapezium $PQRS =$ area of rect PTRS + area of triangle TQR $= (PS \times SR) + (\frac{1}{2} \times TQ \times TR)$ $= (h \times l_2) + [\frac{1}{2} \times (l_1 - l_2) \times h]$ using result of (a) $= h l_2 + \frac{1}{2} h(l_1 - l_2)$ $= h l_2 + \frac{1}{2} h l_1 - \frac{1}{2} h l_2$ $= \frac{1}{2} h l_1 + h l_2$ $= \frac{1}{2} h(l_1 + l_2)$	1 1 1	MS-M1.2/ MS11-4/ MS-A1/ MS11-10	6
(c)	Let the area = 250 and $(l_1 + l_2) = 25$ Then, $250 = \frac{1}{2} \times h \times 25$ $25h = 500$ $h = 20$	1 1	MS-A1/ MS11-10	4/5
(d)	If, $l_2 = 10$ then $l_1 = 15$ and $TQ = l_1 - l_2$ $= 5$	1	MS-A1/ MS11-10	4
40(a)	$V = [(\text{area of outer circle} - \text{area of inner circle})] \times \text{thickness}$ $= [\pi(r_1)^2 - \pi(r_2)^2] \times 2 \text{ cm}$ $= \pi[(\frac{d_1}{2})^2 - (\frac{d_2}{2})^2] \times 2$ $= \pi[\frac{d_1^2}{4} - \frac{d_2^2}{4}] \times 2$ $= \frac{2\pi}{4} [d_1^2 - d_2^2]$ $= \frac{\pi}{2} [d_1^2 - d_2^2]$	1 1 1	MS-M1.2/ MS11-10/ MS-A1/ MS11-10	6
(b)	Let $V = 18\pi$, $d_2 = 8$ Then, $18\pi = \frac{\pi}{2} [d_1^2 - 64]$ $36 = d_1^2 - 64$ $d_1^2 = 100$ $d_1 = 10 \text{ cm}$	1 1	MS-A2/ MS11-10	5